

A blue triangle graphic pointing to the right, located to the left of the main title.

FA report of Abnormal Display issue

JUNGLIM G140HAB01.2 VLD

Problem Statement



- General Information

Brand/Project	JUNGLIM				Defect Phenomenon		V LINE Defect															
ES	YH Chuang				Defect Rate or Q' ty		1pcs															
AUO Model	G140HAB01.2				Defect Station		FIELD RETURN															
AUO S/N	DF5580C6TRZZZS0100				RMA No.		702610101420															
Beol/Module	<table><tr><th>ARRAY_FAB_ID</th><th>FEOL_FAB_ID</th><th>CF_FAB_ID</th><th>BEOL_FAB_ID</th><th>LCM_BEOL_FAB_ID</th><th>JI_FAB_ID</th><th>MODULE_FAB_ID</th></tr><tr><td>L5B</td><td>L5B</td><td></td><td>L5B</td><td>S01</td><td></td><td>S01</td></tr></table>								ARRAY_FAB_ID	FEOL_FAB_ID	CF_FAB_ID	BEOL_FAB_ID	LCM_BEOL_FAB_ID	JI_FAB_ID	MODULE_FAB_ID	L5B	L5B		L5B	S01		S01
ARRAY_FAB_ID	FEOL_FAB_ID	CF_FAB_ID	BEOL_FAB_ID	LCM_BEOL_FAB_ID	JI_FAB_ID	MODULE_FAB_ID																
L5B	L5B		L5B	S01		S01																
Week Code	25/44				Issue Date		2026.05.08															

- Defect Panel S/N Picture



Problem Statement

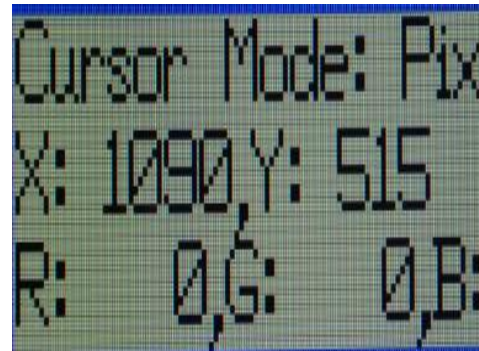
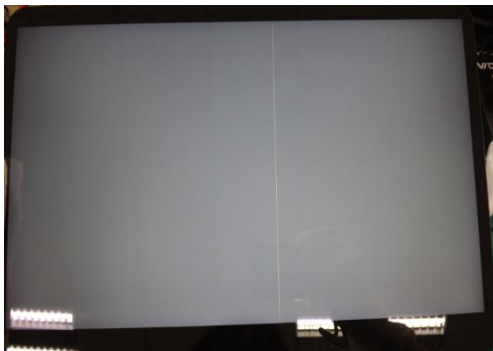


- Describe the phenomenon at customer side

1. V-line defect observed at customer side.

- Defect Phenomenon

1. A vertical line (V-line) is visible after display activation.
2. The defect location is identified at approximately $X = 1090$.



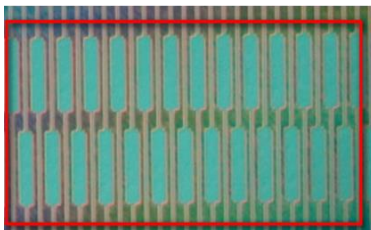
Root Cause Analysis

Step 1: Line Tracing Inspection:

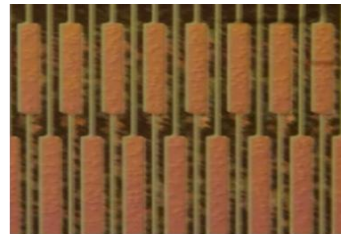
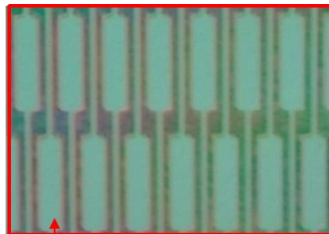
Abnormality was found at the X2 IC bonding area along the V-line path.

Findings:

1. Local bonding line shows poor compression contact.
2. Compared to the normal region, the affected area shows unclear conductive particle distribution, indicating insufficient electrical connection.



X2 IC Bonding poor bonding



Normal region



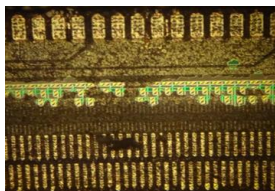
Root Cause Analysis

Step 2: IC Removal Inspection

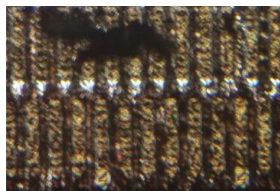
After removing the X2 IC, no abnormality is observed on both the cell side and IC side bonding pads.

Findings:

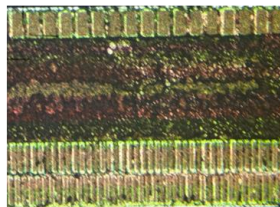
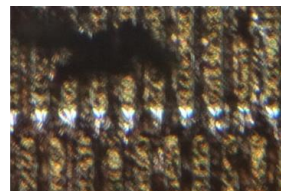
1. Cell-side electrode pattern is intact without damage or open trace.
2. IC-side pads show normal morphology.
3. No contamination, residue, or mechanical defect is found after IC removal.



After Remove IC



Cell side



IC side

Conclusion



1. The V-line defect is attributed to **abnormal bonding quality at the X2 IC interface**.
2. Non-uniform compression at the bonding area may result in **unstable electrical contact**, leading to signal failure on the corresponding channel.
3. After IC removal, both **cell-side and IC-side pads are confirmed to be normal**, ruling out design or material-related concerns.

→ The root cause is determined to be **bonding process variation**, resulting in insufficient contact at the X2 IC bonding region.

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The image features a large blue trapezoidal shape on a white background. The trapezoid has a lighter blue top edge and a darker blue bottom edge, with a white diagonal line on the left side. Centered within the trapezoid is the logo 'NUO' in a bold, white, sans-serif font. Below the logo, the tagline 'Innovating Life' is written in a smaller, white, italicized sans-serif font.

NUO

Innovating Life